

# STAREHE BOYS' 2025 PREDICTION



Kenya Certificate of Secondary Education  
**ANTICIPATED KCSE EXAM**



233/3

**Chemistry (Practical)**

**Paper 3**

Time: 2¼ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

## INSTRUCTIONS:

- Answer **ALL** questions in the spaces provided.
- You are **NOT** allowed to start working with the apparatus for the first 15 minutes of the 2 ¼
- Hours allowed for this paper. This time will enable you read through the question paper and make sure you have all the chemicals and apparatus required.
- Mathematical tables and electronic calculators may be used
- All working must be clearly shown where necessary.

## FOR EXAMINER'S USE ONLY.

| Question    | Maximum score | Candidate's score |
|-------------|---------------|-------------------|
| 1           | 20            |                   |
| 2           | 12            |                   |
| 3           | 8             |                   |
| Total score | 40            |                   |

Answer **ALL** questions in the spaces provided.

1. You are provided with:

- Solution A of Potassium manganate (VII).
- 0.05M solution B of oxalic acid.
- Solution C containing 4.9g of ammonium iron (II) Sulphate,  $(\text{NH}_4)_2 \text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$ , in  $250\text{cm}^3$  of water.
- You are required to:
  - i) Determine the rate of reaction between oxalic acid and Potassium manganate (VII).
  - ii) Standardize the solution A.

**PROCEDURE I:**

Using a measuring cylinder, place  $1\text{ cm}^3$  of solution A into each of the five (5) test-tubes in a rack.

Clean the measuring cylinder and use it to place  $19\text{cm}^3$  of solution B into a boiling tube.

Prepare a water bath by placing about  $200\text{cm}^3$  of water into a beaker and start to heat. Place a thermometer into solution B and place it in the warm water until it attains a temperature of  $40^\circ\text{C}$ .

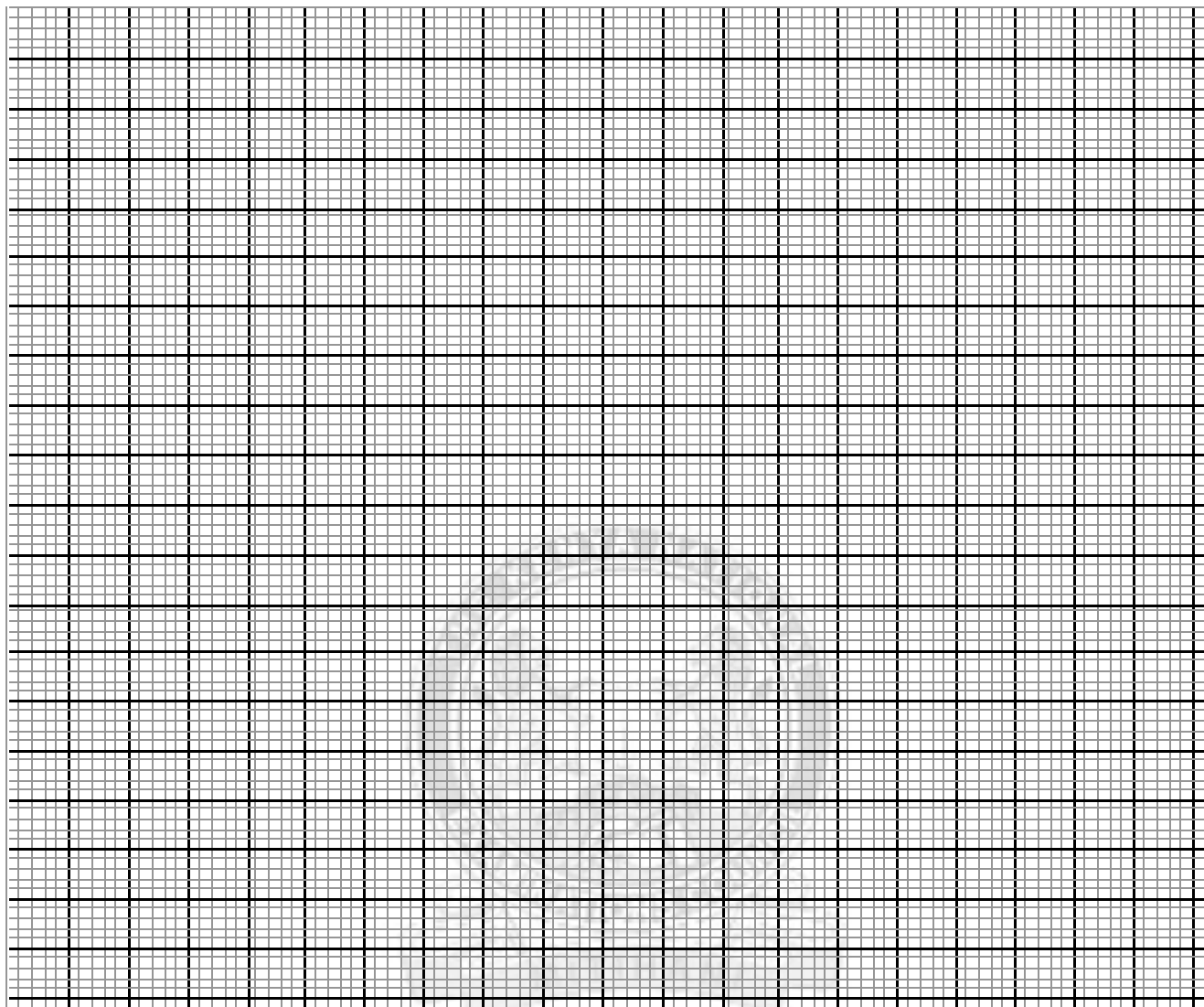
Remove the boiling tube from the water-bath and place it in the test-tube rack. Add the first portion of solution A immediately and at the same time start a stop watch. Record the time taken of solution A to be decolourised in table I below. Repeat the procedure at temperatures of  $50^\circ\text{C}$ ,  $60^\circ\text{C}$ ,  $70^\circ\text{C}$  and  $80^\circ\text{C}$  to complete the table.

| Temperature of solution B ( $^\circ\text{C}$ ) | 40 | 50 | 60 | 70 | 80 |
|------------------------------------------------|----|----|----|----|----|
| Time taken for decolourisation (secs)          |    |    |    |    |    |
| $1/t\text{ sec}^{-1}$                          |    |    |    |    |    |

**(3 marks)**

i) Plot a graph of  $1/t$  against temperature (X-axis)

(3 marks)



ii) From the graph determine the time taken for the mixture to decolourise at 65°C (2 marks)

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iii) How does the rate of reaction between oxalic acid and Potassium manganate (VII) vary with temperature? (1 mark)

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## **PROCEDURE II**

Fill a burette with solution A. Pipette 25cm<sup>3</sup> of solution C into a conical flask and titrate the solution A against solution C until a permanent pink colour just appears. Record your results in table II below and repeat the procedure to fill the table.

|                                              | I | II | III |
|----------------------------------------------|---|----|-----|
| Final burette reading (cm <sup>3</sup> )     |   |    |     |
| Initial burette reading (cm <sup>3</sup> )   |   |    |     |
| Volume of solution A used (cm <sup>3</sup> ) |   |    |     |

(4 marks)

i) Determine the average volume of A used .....cm<sup>3</sup>

(Show how you arrive at your answer)

(1 mark)

ii). Calculate the concentration of solution C in moles per litre (Fe, = 56, S = 32,

O = 16, N = 14, H = 1)

(2 marks)

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iii) Find the number of moles of solution C used.

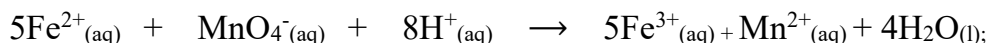
(1 mark)

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iv) Given the ionic equation for the reaction is



Find the number of moles of solution A used.

(1 mark)

**(2 marks)**

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2. You are provided with solid Q. Carry out the tests below and record your observations and inferences in the table below.

i) Place half a Spatula full of solid Q in a clean dry test-tube and heat gently then Strongly.

| Observations                                                                                        | Inferences      |
|-----------------------------------------------------------------------------------------------------|-----------------|
|  <p>(1 mark)</p> | <p>(1 mark)</p> |

ii). Place the remaining solid Q in a boiling tube and add about 5cm<sup>3</sup> of distilled water and shake well. Divide the resulting mixture into four portions for the tests below.

| Observations | Inferences |
|--------------|------------|
|--------------|------------|

|          |          |
|----------|----------|
| (1 mark) | (1 mark) |
|----------|----------|

a). To the first portion add Sodium hydroxide solution dropwise until in excess.

| Observations | Inferences |
|--------------|------------|
| (1 mark)     | (1 mark)   |

b). To the second portion add 2-3 drops of dilute Sulphuric (VI) acid

| Observations | Inferences |
|--------------|------------|
| (1 mark)     | (1 mark)   |

|                                         |          |
|-----------------------------------------|----------|
|                                         | (1 mark) |
| drops acidified barium nitrate solution |          |
|                                         |          |
|                                         |          |

| Observations | Inferences |
|--------------|------------|
| (1 mark)     | (1 mark)   |

**d).** To the fourth portion add 2-3 drops acidified barium nitrate solution

| Observations | Inferences |
|--------------|------------|
| (1 mark)     | (1 mark)   |

3. You are provided with solid L. carry out the tests below on L and record the observations and inferences in the spaces provided.

(a) Place half of solid L in a boiling tube and add about 5cm<sup>3</sup> of distilled water

(i)

| Observations | Inferences |
|--------------|------------|
| (1 mark)     | (1 mark)   |

Divide the solution into two portions and carry out the tests below.

(ii). To the first portion add 2-3 drops of acidified potassium manganate (VII).

| Observations    | Inferences      |
|-----------------|-----------------|
| <p>(1 mark)</p> | <p>(1 mark)</p> |

**(iii). To the second portion add Sodium carbonate provided (1 mark)**

| Observations | Inferences |
|--------------|------------|
|              |            |



|          |          |
|----------|----------|
| (1 mark) | (1 mark) |
|----------|----------|

(b). Place the remaining solid L in metallic spatula and ignite it.

| Observations | Inferences |
|--------------|------------|
| (1 mark)     | (1 mark)   |

